

- 2 The level of sound produced by a particular type of machine was measured for a random sample of 11 such machines. The results, in suitable units, are shown below.

Machine	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>
Sound level	7.66	8.48	8.21	7.98	8.01	7.77	8.25	8.11	8.03	8.16	7.92

- (a) Use a Wilcoxon signed-rank test to test whether the average sound level produced by this type of machine is more than 8.00. Use a 5% significance level. [6]

This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b) Give a reason why a Wilcoxon signed-rank test may be more appropriate than a t -test in this case. [1]

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